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Effects of Kindergarten Retention Policy on Children's Cognitive Growth in Reading and Mathematics

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Grade retention has been controversial for many years, and current calls to end social promotion have lent new urgency to this issue. On the one hand, a policy of retaining in grade those students making slow progress might facilitate instruction by making classrooms more homogeneous academically. On the other hand, grade retention might harm high-risk students by limiting their learning opportunities. Analyzing data from the US Early Childhood Longitudinal Study Kindergarten cohort with the technique of multilevel propensity score stratification, we find no evidence that a policy of grade retention in kindergarten improves average achievement in mathematics or reading. Nor do we find evidence that the policy benefits children who would be promoted under the policy. However, the evidence does suggest that children who are retained learn less than they would have had they instead been promoted. The negative effect of grade retention on those retained has little influence on the overall mean achievement of children attending schools with a retention policy because the fraction of children retained in those schools is quite small. Nevertheless, the effect of retention on the retainees is considerably large.

Keywords: *causal inference, grade retention, kindergartners, math growth, propensity score, principal stratification, reading growth*

A BASIC feature of modern schooling is the assignment of children to classrooms composed of those who are relatively homogenous in age. Progress in schools is marked by transitions through age-based grade levels. However, not all children progress at the same rate academically or socially, which poses a question about how to manage the growing inequality of developmental status as kids grow older. One option is to adopt a policy of grade retention. Under this

policy, students showing weak academic growth are retained in their current grade rather than being promoted to the next grade. This policy increases heterogeneity of age within classrooms but it may also reduce heterogeneity of academic achievement within those classrooms, which may arguably make instruction more efficient. Opponents of grade retention have long advocated instead a policy of “social promotion,” under which all students are promoted to

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maintain homogeneity of age within classrooms. Under the social promotion policy, remedial instruction in a later grade could possibly be designed to overcome deficiencies in academic growth exhibited in early grades.

In recent years, “ending social promotion” has become a popular slogan within the movement to set and maintain high standards and educational accountability. In this new climate, many schools have adopted grade retention at most grade levels, and even in kindergarten (Ellwein & Glass, 1989; Hauser, 1998; Roderick, Bryk, Jacobs, Easton, & Allensworth, 1999). Based on the National Household Education Survey, Zill, Loomis, and West (1997) reported that the kindergarten retention rate was about 6% in 1993 and 5% in 1995. Nonetheless, many school districts and schools continue to consider retention, especially kindergarten retention, to be an inappropriate and ineffective remedial device that should be eliminated. Rigorous inquiry into the effects of kindergarten retention is, therefore, timely. We focus this study on evaluating the causal effects of the kindergarten retention policy on children’s cognitive growth in reading and mathematics.

One important question concerns the effect of adopting a grade retention policy on school-average academic achievement. A widely endorsed argument is that, when low-achieving students are retained in grade, the academic status of children in a classroom will become more homogeneous, easing the teacher’s task of managing instructional activities (Byrnes, 1989; also see Shepard & Smith, 1988, for a review). In particular, retaining some children in kindergarten may allow the first-grade teacher to teach at a higher level, benefiting those who would be promoted under the policy. Meanwhile, children who view grade retention as a punishment may study harder to avoid being retained in the future. Some have argued that, in comparison with the social promotion policy, repeating a grade is perhaps developmentally more appropriate and may make learning more meaningful for children who are struggling (Plummer & Graziano, 1987; Smith & Shepard, 1988). If these arguments are correct, adopting a policy of grade retention will benefit those promoted and those retained, thus boosting achievement overall.

However, the proposition that grade retention will help the retainees is particularly contentious.

The likely impact of a grade retention policy on those retained has dominated past debate and research on grade retention policies. This preoccupation is justifiable in that children likely to be retained are at high risk of later difficulty, including low grades, low test scores, and dropping out of school prior to high school graduation (Rumberger & Larson, 1998), of special concern for policy. If grade retention helps these students, few would disagree with the retention policy. After all, those promoted might benefit as well and would not likely be hurt by such a policy. However, if grade retention impedes the academic growth of the retained children, a trade-off may arise, depending on whether those who are promoted benefit from the policy. How such a trade-off might be managed depends not only on one’s values but also on the size of the effects on the two groups. It makes sense, therefore, to evaluate the impact of the policy overall as well as its effects on those who are promoted and those who are retained.

What Do We Know About the Effects of Grade Retention?

Is retention a solution to the academic problems experienced by young students? From a developmental perspective, many educators and parents believe that children who are behind their peers socially or cognitively need more time to become mature before progressing to the first grade (Smith & Shepard, 1988). In contrast, some developmental psychologists argue that grade retention stifles young children’s cognitive and social development (Morrison, Griffith, & Alberts, 1997). Similar controversies exist with regard to the effects of grade retention on children’s emotional well being. While some expect that changing the peer reference group will increase the self-esteem of those kindergarten retainees (Plummer & Graziano, 1987), others claim that retention stigmatizes retainees, leading to lower teacher, parent, and self-expectations (Jackson, 1975; Shepard, 1989).

Contrasting stances on retention policy tend to reflect broad disagreements about curriculum and instruction. One view is that retention will benefit the students who are not equipped to learn higher-level material because they have not yet mastered the lower-level facts and skills. In contrast, social constructivist theories hold that all children bring considerable knowledge and abil-

ities that enable them to construct new knowledge. Arguably, retention would be unnecessary in a classroom environment capable of mobilizing these resources during instruction (Peterson, 1989). If so, reforming instruction to correct problems would be more effective than simply having children repeat an unsuccessful experience (Karweit, 1992; Leinhardt, 1980; Reynolds, 1992; Tanner & Galis, 1997).

Comprehensive reviews of almost a century of empirical research on grade retention have not resolved this controversy (Holmes, 1989; Jackson, 1975; Jimerson, 2001; Shepard, 1989). A large number of analyses show negative results of retention on children's academic achievement and personal-social development. Meanwhile, a similarly large number of studies report no statistically significant differences in the outcomes between the retained and the promoted groups. Results of a much smaller number of studies favored retention. While the weight of evidence seems to suggest that the impact of retention on those retained is harmful, methodological disagreements have fostered continued uncertainty.

Past studies of retention have used two primary strategies: (a) same-grade comparisons, and (b) same-age comparisons. The "same-grade" studies, which constitute the majority of all retention studies, compared the outcomes of children repeating a grade with those of children experiencing the grade for the first time. The "same-grade" study design is useful for addressing research questions other than those about the causal effects of retention. Researchers are able to compare the standing of retainees relative to their classmates before and after the retention, but are unable to provide information on the academic progress the retained children might have displayed had those children been promoted. Results from these studies have been mixed regarding the average difference in the outcomes between the retained group and the comparison group (Alexander, Entwisle, & Dauber, 2003; Ferguson, 1991; Jimerson, 1999; Jimerson, Carlson, Rotert, Egeland, & Sroufe, 1997; Karweit, 1999; Mantzicopoulos & Morrison, 1992; McCoy & Reynolds, 1999; Meisels & Liaw, 1993; Peterson, DeGracie, & Ayabe, 1987; Reynolds, 1992; Shepard & Smith, 1989).

Studies based on "same-age" comparisons, in contrast, compare the outcomes of retained chil-

dren with those of same-age children who were, in fact, promoted to the next grade. Results from the "same-age" studies have been more consistent than those from the "same-grade" studies. Researchers have most often found that the promoted children display better outcomes than the same-age retained children, even after adjustment for some prior background characteristics (Alexander et al., 2003; Ferguson, Jimerson, & Dalton, 2001; Holmes, 1989; Karweit, 1999; McCoy & Reynolds, 1999; Meisels & Liaw, 1993; Reynolds, 1992). When the comparison group was restricted to the same-age poor-performing peers, the results still showed that the retained students tend to lag behind after being held back for 1 or more years (e.g., Alexander et al., 2003; Karweit, 1999).

The Challenge of Causal Inference

Not surprisingly, given the impracticality of using large-scale experimental designs to study retention effects, most previous empirical studies have relied on nonexperimental data. While supplying useful information about the expected academic progress of subgroups of students, these studies are nonetheless troublesome from the standpoint of causal inference. First, most promoted students have little or no risk of ever being retained. Such "low-risk" students, therefore, provide little or no information about the academic progress expected of the retained students had those "high-risk" students been promoted (Holland, 1988). Second, statistical adjustment for a limited number of background variables, by far the most common analytic strategy employed, cannot be relied upon to remove bias when the two "treatment groups" are barely comparable. Predictions about what might happen to high-risk students if promoted are based largely or in part on "extrapolations" that use a linear model assumption to compensate for the lack of a viable comparison group (Little, An, & Johanns, 2000; Lord, 1967; Shadish, Cook, & Campbell, 2002). Finally, adjustment by means of a linear model sharply constrains the number of background characteristics that can be controlled. In general, the validity of the conclusions drawn from the statistical results of these studies relies heavily on strong but unexamined assumptions. These include the assumption that all confounding variables have been measured, and the assumption that the variation in the outcomes can

be explained by linear and additive models (Little, 1985; Little et al., 2000; Stone, 1993).

Research Questions

Previous retention studies have focused almost exclusively on retention as an individual-level treatment for retainees. The weight of evidence suggests that those who are retained exhibit less growth than they would have had they been promoted. However, this proposition remains contentious, in part because valid causal inference is challenging. Meanwhile, there has been scant evidence regarding the average effect of the retention policy especially on the promoted children. For example, Smith and Shepard (1988) found remarkably less emphasis on instructional adaptation in high-retaining schools in one school district, suggesting a negative impact of the retention policy on student learning. This hypothesis remains yet to be examined with nationally representative samples. To achieve a more comprehensive understanding of the empirical issues underlying the policy debate, we go beyond examining the effect of retention on those retained to consider the average effect of adopting a retention policy on all students and also the effect of such a policy on those who would be promoted if the policy were adopted. Asking all these questions is important in light of possible trade-offs that will arise if the policy affects different subgroups differently. Our empirical questions, then, are three:

1. What is the average effect of the kindergarten retention policy? To be more explicit, should we expect to see a change in children's average learning outcomes if a school changes its retention policy?
2. What is the average impact of a school's retention policy on children who would be promoted if the policy were adopted? To answer this question, we adopt the method of principal stratification developed by Frangakis and Rubin (2002).
3. What is the effect of kindergarten retention on those who are retained? Our aim here is to estimate how much more or less kindergarten retainees would have learned, on average, had they been promoted to the first grade rather than retained.

If we find positive effects of the retention policy overall, positive effects on those who would be promoted under the policy and positive effects of being retained, a strong case can be made in

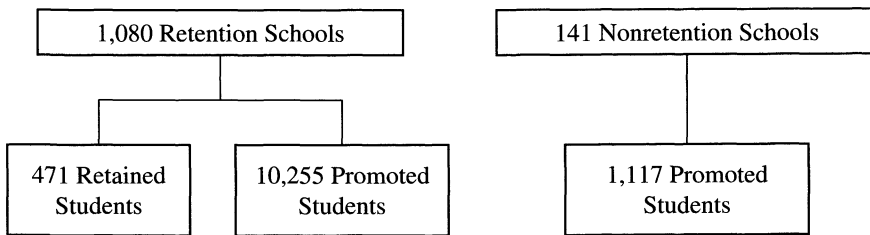
favor of adopting the policy. If all three effects are negative, the case against the retention policy is strengthened. If, on the other hand, the results are mixed, the policy implications may be less clear, depending on the size of the effects.

Data

We selected data from the Early Childhood Longitudinal Study Kindergarten cohort (ECLS-K) released by the US National Center for Education Statistics (NCES). The data set contains repeated observations of a nationally representative sample of students, their families, teachers, and schools over the kindergarten and first-grade years from Fall 1998 to Spring 2000. The total sample includes more than 20,000 kindergartners. Most of the students were observed in the fall and the spring of the kindergarten year and then in the spring of the treatment year. In addition, a random subsample of students was observed in the fall of the treatment year. Among the first-time kindergartners in the ECLS-K sample, retention/promotion status in the treatment year was recorded for 13,520 students who had at least one reading or math test score during the treatment year. In addition, 1,221 schools provided information on their policies regarding kindergarten retention during the treatment year. Because information about school retention policy was missing for 1,677 students, our analytic sample, as illustrated in Figure 1, includes 471 kindergarten retainees and 10,255 promoted students in 1,080 retention schools, and 1,117 promoted students in 141 nonretention schools.

A comparison between the 11,843 children in our analytic sample and the more than 20,000 children in the full sample suggested some differences in demographic composition. In general, in comparison with the full sample, the group with treatment information represented a lower proportion of poor and minority children who were less likely to have come from non-English-speaking families (see Table 1). As a result, the schools in our analytic sample tended to have a lower percentage of minority and free-lunch students than the schools in the full sample.

The outcome variables are reading and math ECLS-K scale scores calibrated via item response theory (IRT) (Hambleton, Swaminathan, & Rogers, 1991). In the data set available to us, the test scores of each subject obtained from up to four repeated assessments over the 2 study

FIGURE 1. *ECLS-K sample of students and schools in each treatment group.*

years have been equated on the same scale. This enables researchers to assess the reading and math growth of each student over time, and to compare the reading and math achievement, respectively, of students from different grade levels. See Table 2 for descriptive information about the four rounds of reading and math scale scores.

Causal Inference Theories and Methods

Rubin's conceptual framework of causal inference (Holland, 1986; Rubin, 1978) clearly defines the causal effect of one treatment versus another for each unit as the difference between the two potential outcomes corresponding to the two treatments. This causal model was presented under the stable unit treatment value assumption (SUTVA) that one's potential outcome value associated with each treatment does not depend on how the treatments are assigned and what treatments are received by other units (Rubin, 1986). In the multilevel setting of schools, we expect that this assumption will not hold. Rather, potential outcomes will depend on the school one attends as well as the peers attending that school. In principle, potential outcomes will depend on classroom assignment as well, though the ECLS-K data on retention are too sparse to support class-

room-level analysis. Hong (2004) and Hong and Raudenbush (2003) have shown that SUTVA can be relaxed to incorporate multilevel data that arise in education by identifying the level at which a treatment is implemented and representing the effects of persons, classes, and schools explicitly in the model. Because our data are nonexperimental, we combine multilevel modeling (Raudenbush & Bryk, 2002) and propensity score stratification (Rosenbaum & Rubin, 1983) to adjust for selection bias.

Our evaluation of the kindergarten retention policy involves two sets of treatments. First, each school is characterized by a binary indicator of its retention policy, with $D = 1$ if the school does allow retention in kindergarten and $D = 0$ if the policy forbids the retention of children in kindergarten. Our first causal question is: What is the average effect of the school's kindergarten retention policy? This requires that we estimate the average causal effect of $D = 1$ versus $D = 0$. Of course schools are not assigned at random to these policies, and a variety of school characteristics may influence the choice of policy. Through propensity score stratification, we were able to balance the 141 nonretention schools and the 1,080 retention schools on

TABLE 1
Comparison Between ECLS-K Full Sample and Analytic Sample

Student demographic features	Full sample	Analytic sample
Average age	5.54	5.53
Proportion of girls	0.49	0.49
Proportion of white	0.55	0.61
Proportion of black	0.15	0.13
Proportion of Hispanic	0.18	0.15
Proportion of Asian	0.06	0.05
Proportion below poverty	0.19	0.17
Proportion speaking non-English at home	0.15	0.11

TABLE 2
Descriptive Statistics of the Four Rounds of Assessment

A. Reading			
<i>Reading IRT scale score</i>	<i>n</i>	<i>M</i>	<i>SD</i>
Fall, kindergarten year	11,121	23.35	8.72
Spring, kindergarten year	11,450	33.56	10.90
Fall, treatment year	3,449	39.18	12.76
Spring, treatment year	11,797	56.28	13.53

B. Mathematics			
<i>Math IRT scale score</i>	<i>n</i>	<i>M</i>	<i>SD</i>
Fall, kindergarten year	11,506	20.16	7.18
Spring, kindergarten year	11,652	28.36	8.65
Fall, treatment year	3,481	33.38	9.28
Spring, treatment year	11,795	43.78	8.95

238 school-level characteristics that we regarded as pretreatment covariates.

Our second treatment is characterized by the binary indicator $Z = 1$ if a child is retained in kindergarten and $Z = 0$ if that child is promoted to the first grade. Clearly, Z will always be zero when a school bans retention. The average causal effect of $Z = 1$ versus $Z = 0$ corresponds to our third question regarding the average effect of being retained on those who were retained. To answer this question, we use data from the schools that allow retention.¹ We use propensity score stratification to balance the 471 retained students with the promoted students attending schools that allow retention on 207 pretreatment covariates.

Understanding the average causal effect of Z on those attending schools that allow retention is extremely important because intense interest focuses on how retention affects those retained, a high-risk population of great concern to educators. However, knowing such an effect does not inform us about the consequences of the policy for those who would be promoted if the policy were adopted. And a sensible policy decision might depend on the direction and magnitude of such an effect.

Estimating the causal effect of the retention policy on those who would be promoted under the policy is tricky because we cannot observe which children attending schools that ban retention would have been promoted if the policy were changed. In the language of Frangakis and Rubin (2002), we have two principal strata: those who are at risk of retention if the policy were adopted (principal stratum 1) and those who would almost certainly be promoted if the policy were adopted (principal stratum 2), as shown in Table 3. Children in the first principal stratum possess three potential outcomes: $Y(D = 0, Z = 0)$ is the outcome a child would display if the school were to ban retention; $Y(D = 1, Z = 0)$ is the outcome a child would display if that child's school allowed retention and if that child were promoted; while $Y(D = 1, Z = 1)$ is the outcome the child would display if the child were, in fact, retained. The average difference between the latter two potential outcomes, $E[Y(D = 1, Z = 1) - Y(D = 1, Z = 0)]$, is the average causal effect of grade retention on children who would be retained in schools that allowed retention. Those in the second principal stratum possess only two potential outcomes: $Y(D = 1, Z = 0)$, the outcome

TABLE 3
Principal Stratification and Potential Outcomes

Principal stratum	Potential outcome			Causal effect
1	$Y(D = 0, Z = 0)$	$Y(D = 1, Z = 0)$	$Y(D = 1, Z = 1)$	$E[Y(D = 1, Z = 1) - Y(D = 1, Z = 0)]$
2	$Y(D = 0, Z = 0)$	$Y(D = 1, Z = 0)$		$E[Y(D = 1, Z = 0) - Y(D = 0, Z = 0)]$

that a child would display if the school allowed retention but promoted that child; and $Y(D = 0, Z = 0)$, the outcome that the child would display if the school banned retention. The average difference between these two potential outcomes, $E[Y(D = 1, Z = 0) - Y(D = 0, Z = 0)]$, is the quantity we seek in posing our question about the average causal effect of the retention policy on those who would be promoted if the policy were adopted.

The problem, of course, is that we do not observe membership in the two principal strata. In particular, we do not observe which students attending schools that ban retention would be promoted if the school were to allow retention. We addressed this problem as follows. First, we predicted the probability of grade retention using data from those who attended schools that allowed grade retention. Specifically, we used a subset of the 207 available covariates to make this prediction. This enabled us to identify 3,087 students whose risk of retention was essentially null. This group constitutes 30.1% of all students attending schools that allowed retention. Keeping in mind that only 4.6% of the students in those schools were actually retained and that many powerful covariates were available to predict retention, it is quite safe to say that this group had essentially no risk of being retained. We then applied the same prediction equation to those students attending schools that banned retention, identifying 820 students with similarly near-null predicted probabilities of retention. We could estimate the average causal effect of the retention policy on those who would be promoted by comparing the 3,087 low-risk students in the retention schools to the 820 low-risk students in the nonretention schools after balancing all the school-level pretreatment covariates.

Results

Predictors of Kindergarten Retention Policy

Recall that to answer our first question (What is the average effect of the retention policy?) requires that we balance two treatment groups $D = 1$ (those that allow retention) and $D = 0$ (those that ban retention) on all known pretreatment confounders. We, therefore, modeled the selection of schools into the retention policy.

Using the ECLS-K data set, we identified 238 school-level pretreatment covariates that displayed

a bivariate association with a school's kindergarten retention policy. In general, the retention schools seemed to have notable advantages over the nonretention schools. The kindergarten retention policy tended to be adopted by nonpublic schools, suburban schools, schools with a comparatively low percentage of minority students, and schools with a higher percentage of white teachers. On average, the retention schools had smaller kindergarten class sizes, more parent involvement, and better order in classrooms, schools, and communities. Kindergartners in retention schools had a higher average reading test score in the pretreatment year, compared with those in nonretention schools.

Predictors of Kindergarten Retention

To answer our second and third questions required that we predict the probability of grade retention in schools that allow grade retention. Through bivariate analyses, we identified 207 pretreatment covariates available in the ECLS-K data that were significantly associated with kindergarten retention. Predictors include a child's demographic and psychological characteristics, along with family, classroom, and school characteristics.

In general, a child was more likely to be retained if the child was non-Hispanic, male, from a family with lower socioeconomic status, if the child had never received center-based care, and if the child had entered the kindergarten at a younger age. Moreover, children from single-parent families with more siblings were more likely to be retained. So were children with a disability or having fallen 2 or more weeks behind in schoolwork because of health problems during the kindergarten year. Children destined to be retained in kindergarten tended to have parents who showed less commitment to parenting responsibilities. Parents of retained children tended to have lower educational expectations of their children, according to the measures collected at the beginning of the kindergarten year. These families were less likely to send their children to extracurricular activities. They usually had no computer for child use, and had fewer books for their children to read outside of school.

A child's cognitive and social developmental status at the beginning and the end of the kindergarten year seemed to be major determinants of the retention decision. Children who were to be

retained consistently showed lower achievement status in literacy, mathematics, and general knowledge on all types of assessments. Moreover, these children tended to receive a lower rating from their teachers and parents on approaches to learning, self-control, and inter-personal skills, and were rated higher on emotional or behavioral problems.

A teacher's perception of her students' readiness for school may affect, to some extent, her choice of learning opportunities for the class, which may, in turn, affect the children's learning results and, consequently, the treatments that they will receive in the following year. The results showed that children who would be retained at a later time were more likely to be placed in the lowest reading group in the kindergarten class. Such children almost never had a chance of moving to a higher-level group. Children destined for retention were more likely to receive individual tutoring or pull-out, small-group instruction in reading and mathematics during the kindergarten year. Still, these children were often judged by their teachers to be not working at their best ability.

The composition of one's kindergarten class, along with teacher characteristics, shapes the learning environment during a child's kindergarten year and predicts the child's probability of being retained later. The following kindergarten classroom characteristics had positive bivariate associations with grade retention: proportion of boys, proportion of 3- or 4-year-olds, proportion of children currently repeating the kindergarten, and proportion of children with disabilities. Children destined to be retained tended to be in a kindergarten class with a smaller number of children recognizing letters at the start and/or being classified as gifted. They were more likely than promoted children to be in a class with a smaller proportion of Hispanic and LEP students. In addition, the teacher of such a kindergarten class was more likely to report frequent behavioral problems at the beginning of the year. Teachers of children to be retained tended to spend less time in reading and literacy instruction; they tended to cover lower-level content topics in reading, literacy, and mathematics. The teacher in classes of retained children tended to hold different standards based on children's capability and tended to emphasize the importance of home-assisted kindergarten learning. Moreover, these kindergarten classes were more likely to be taught

by teachers who did not have an educational degree beyond the Bachelor's, by those with less experience in teaching kindergarten and first grade, and by those with less experience in teaching ESL and bilingual education.

Finally, the schools that the kindergarten retainees-to-be attended during their first kindergarten year were characteristic of a smaller size, being nonpublic, with fewer classroom teachers and ESL-bilingual teachers, a lower percentage of Hispanic children, inadequate instructional resources and facilities, lower teacher salary, lower daily attendance, no special services for children with a disability, fewer attempts at outreach to parents, more security measures, and a lower safety rating. These schools tended to have principals who had spent more years teaching preschool and kindergarten and fewer years teaching higher grade levels, had taken more courses in early childhood education or child development and fewer courses in administration and ESL, and had spent more hours per week teaching and fewer hours with required paperwork. According to the reports from these principals, their schools put less emphasis on goals of language and number skills and more emphasis on behavioral goals, were less open to new ideas and methods, and had been less successful in providing help to low achievers. These principals were less likely to consider the evaluation of their performance to depend upon their students' performance.

The above profile suggests that the pretreatment experiences of the kindergarten retainees were different, on many aspects, from the experiences of their promoted peers. These pretreatment differences were potential sources of selection bias that we must take into account in estimating the causal effect of kindergarten retention.

Average Effects of the Kindergarten Retention Policy

On average, students enrolled in the retention schools scored higher than those in the nonretention schools in both reading and mathematics. Near the end of the treatment year, the average test score of the retention school students was 4.02 points higher in reading ($t = 9.488$, $p < .001$) and 2.74 points higher in mathematics ($t = 9.771$, $p < .001$). However, because the retention schools showed some pretreatment advantages over the nonretention schools, we should

not view these unadjusted results as evidence in support of the kindergarten retention policy.

Assuming “strong ignorability,” that is, no unobserved covariates were confounded with treatment given the observed covariates (Rosenbaum & Rubin, 1983), we specified a school’s propensity of adopting the kindergarten retention policy, denoted with Q_j for school j , as a function of the observed school-level pretreatment covariates, W_j .

$$\hat{Q}_j = P(D_j = 1 | W_j). \quad (1)$$

Through stepwise logistic regression, our school-level propensity model finally included 29 covariates and the quadratic terms for 4 of these 29 variables.² We then divided the sample of nonretention schools into seven strata on the basis of the logit of the school-level propensity score, and identified the retention schools within each stratum (see Table 4). No matches were found for the 10 nonretention schools that were least likely to adopt the kindergarten retention policy. These 10 schools would not contribute to the estimation of the kindergarten retention policy effects. Distribution balance was achieved in the remaining strata in about 96% of more than 200 pretreatment covariates. The within-stratum difference in about 4% of the pretreatment covariates could largely be attributed to the Type I error related to sampling fluctuation.

We next examined the effects of the kindergarten retention policy on the reading and math outcomes across the propensity strata. The results are displayed in Tables 5 and 6. The within-stratum policy effects ranged from -4.00 to 4.96 in reading, and from -3.83 to 3.78 in mathematics. On average, students attending retention schools did not seem to be better off in reading and math

learning when compared with their counterparts in the nonretention schools.

Observing no systematic variation in the retention policy effects across the strata, we adopted a model-based approach to estimating the average retention policy effect, δ_D . Here we combined propensity score stratification with covariance adjustment for the logit of the propensity score in order to remove remaining within-stratum bias, if there was any. Below is a two-level hierarchical linear model with a random intercept (Raudenbush & Bryk, 2002, Chapter 2):

$$\begin{aligned} Y_{ij} = & \gamma + \delta_D D_j + \gamma_2 (\text{Dur_F})_{ij} + \gamma_3 (\text{logit_}\hat{Q})_j \\ & + \sum_{h=1}^6 \gamma_{h+3} M_{hj} + u_j + e_{ij}; \\ u_j \sim & N(0, \tau); e_{ij} \sim N(0, \sigma^2). \end{aligned} \quad (2)$$

Here M_{hj} , with $h = 1$ to 6 , are a series of dummy variables indicating six of the seven school-level propensity strata. School j ’s estimated logit of propensity of policy adoption is denoted by $(\text{logit_}\hat{Q})_j$. Because the assessment was not conducted on the same date, $(\text{Dur_F})_{ij}$ indicates the length of time since the beginning of the treatment year when student i in school j took the assessment. In the random part of the model, u_j is the incremental effect of school j to the observed outcome, while e_{ij} is the incremental effect of child i in school j . Some preliminary analyses suggested no variance heterogeneity at either the student level or the school level. Hence, we assumed variance homogeneity for simplicity. After adjusting for selection bias through the above statistical model, the retention effects were estimated to be -0.24 on reading with a standard error of 0.86 , and -0.14 on mathematics with a

TABLE 4
Balance in Logit of the Propensity Score for Kindergarten Retention Policy

Stratum	Retention schools			Nonretention schools		
	<i>N</i>	<i>M</i>	<i>SD</i>	<i>N</i>	<i>M</i>	<i>SD</i>
<i>M</i> = 0	—	—	—	10	−2.34	0.92
<i>M</i> = 1	11	−0.38	0.25	18	−0.57	0.36
<i>M</i> = 2	14	0.12	0.08	14	0.04	0.12
<i>M</i> = 3	113	0.93	0.32	43	0.93	0.35
<i>M</i> = 4	162	1.80	0.23	28	1.81	0.27
<i>M</i> = 5	156	2.45	0.16	14	2.42	0.21
<i>M</i> = 6	624	4.27	2.10	14	3.51	0.68

TABLE 5
Within-Stratum Mean Difference in Reading Achievement between Retention School and Nonretention School Students

Stratum	Retention			Nonretention			Mean diff
	<i>N</i>	<i>M</i>	<i>SD</i>	<i>N</i>	<i>M</i>	<i>SD</i>	
0	0	—	—	154	47.77	13.47	—
1	97	47.88	14.42	153	51.87	10.90	−4.00
2	123	54.73	12.55	73	49.77	14.49	4.96
3	778	51.43	13.53	286	53.12	13.84	−1.69
4	1,350	54.19	13.49	201	52.20	13.48	1.99
5	1,422	56.19	13.43	109	55.53	12.83	0.65
6	6,910	57.98	13.21	141	57.65	13.06	0.33

standard error of 0.55 (see Table 7). We conclude that the average effect of adopting a retention policy is null or very small.

To identify the subpopulation of children at risk of repeating kindergarten, we estimated an individual-level propensity score, denoted by *q*, for the retention school students. The conditional probability that student *i* from pretreatment school *j* would be retained in kindergarten is a function of pretreatment personal and classroom characteristics *X*_{*ij*}, school characteristics *W*_{*j*}, and the residual random effect of the pretreatment school *j*, denoted by *u*^{*}_{*j*}.

$$\hat{q}_{ij} = P\left(Z_i = 1 \middle| D_j = 1, X_{ij}, W_j, u_j^*\right). \tag{3}$$

The student-level propensity model included 39 predictors and seven quadratic terms. We used the logit of the estimated propensity score as an index to indicate the extent to which a child was at risk of repeating kindergarten. A child was considered to be at no such risk if his or her estimated propensity of repetition was so

low that there was no counterpart in the retained group. The kindergarten retainees showed a minimum value of −6.06 in the above index (on a logit scale) corresponding to a predicted probability of retention of .002. There were 3,087 promoted children in retention schools whose logit of the estimated propensity was even lower than this value. These children were identified as at virtually no risk of repetition.

As mentioned earlier, because no child was retained under the nonretention policy, the propensity of kindergarten repetition could not be directly estimated for the nonretention school students. Nonetheless, we could expect that, had a nonretention school changed its policy, some of their students would have been retained in kindergarten. Moreover, we assumed that the nonretention school students would have been subject to some similar selection criteria for retention assignment as approximated by the propensity model in Equation 3. Having estimated the propensity model from the observed data of the retention school students, we used the same model to predict the propensity score for each student enrolled

TABLE 6
Within-Stratum Mean Difference in Math Achievement between Retention School and Nonretention School Students

Stratum	Retention			Nonretention			Mean diff
	<i>N</i>	<i>M</i>	<i>SD</i>	<i>N</i>	<i>M</i>	<i>SD</i>	
0	0	—	—	154	36.70	8.51	—
1	97	37.99	9.42	153	41.82	8.30	−3.83
2	123	42.76	8.21	73	38.98	8.90	3.78
3	778	40.30	9.01	286	41.23	9.06	−0.93
4	1,349	42.21	9.02	201	41.27	9.02	0.94
5	1,421	44.00	8.99	109	43.40	8.29	0.60
6	6,910	44.93	8.65	141	45.50	8.85	−0.57

TABLE 7
Model-Based Estimation of Kindergarten Retention Policy Effects

Population	Reading		Mathematics	
	Coefficient	95% CI	Coefficient	95% CI
All children	-0.24	(-1.93, 1.45)	-0.14	(-1.22, 0.94)
Children at little risk	-0.87	(-3.05, 1.31)	-0.80	(-2.21, 0.61)

in a nonretention school. In this way, we identified 820 nonretention school students who could possibly be at risk of repeating kindergarten under the retention policy, and 297 children who would be at no such risk. We compared the composition of children at no risk of repetition between retention schools and nonretention schools, and found balance in the distribution of more than 98% of the pretreatment covariates.

We then analyzed the model represented in Equation 2 for the subsample of children not at risk of repetition. As shown in Table 7, the policy effects were estimated to be -0.87 on reading and -0.80 on mathematics. Neither was statistically significant. Hence, the kindergarten retention policy showed no effect on these children's reading and math learning. We conclude that the average causal effect of the kindergarten retention policy is negligible on those who are almost certain to be promoted under this policy, subject to the assumption of strongly ignorable treatment assignment, that is, of no unobserved confounding after balancing the strata on the 238 observed school-level covariates.

*Average Effects of Kindergarten
Repetition in Retention Schools*

As we discussed above, an at-risk child attending a nonretention school could have possibly been retained in kindergarten had the school changed its policy. For this reason, the effect of the individual-level retention treatment under the retention policy can be defined for at-risk children in nonretention schools as well as for those in retention schools. However, because all the nonretention school students were promoted, their potential propensity of repetition could not be empirically estimated from the observed data. As a result, the average effects of kindergarten repetition could not be directly estimated for at-risk children attending nonretention schools without overly strong assumptions. Nonetheless, it was

possible to estimate the average effects of kindergarten repetition for at-risk children attending retention schools.

We divided the sample of at-risk students attending retention schools into 15 strata on the basis of the logit of $\hat{q}$ as estimated from Equation 3. Table 8 compares the within-stratum distribution of the logit of this propensity score between the retained group and the promoted group. The eight retainees in the last stratum had no matches in the promoted group. Within each of the remaining 14 strata, the two treatment groups had similar distributions of the logit of $\hat{q}$. The result of hypothesis testing showed no statistically significant between-group difference over all the strata in 97% of 207 pretreatment covariates.

With no statistical adjustment for selection bias, the mean differences between the 471 kindergarten retainees and the 7,168 promoted at-risk students in retention schools were -18.51 in the reading outcome and -11.06 in the math outcome. Given the pretreatment imbalance between the two groups, these mean differences were likely to be negatively biased if viewed as estimates of the causal effects of being retained.

Next, we computed the mean differences in the reading and math outcomes between the retained students and the at-risk promoted students in the retention schools within each individual-level propensity stratum. The results are displayed in Tables 9 and 10. After a year of treatment, the kindergarten retainees generally reached a lower achievement status in both reading and mathematics than did their comparable promoted peers. The within-stratum mean differences in reading ranged from -0.96 to -12.14, while those in mathematics ranged from 1.57 to -7.98.

To generate a more conclusive answer to the causal question, we specified a two-level hierarchical linear model for estimating the average retention effect, δ_z , on the learning outcomes of at-risk children attending retention schools.

TABLE 8
Balance in Logit of the Propensity Score for Kindergarten Repetition in Retention Schools

Stratum	Retention policy					
	Retained			Promoted		
	<i>N</i>	<i>M</i>	<i>SD</i>	<i>N</i>	<i>M</i>	<i>SD</i>
<i>L</i> = 1	9	−5.42	0.40	3,054	−5.38	0.39
<i>L</i> = 2	12	−4.25	0.24	1,661	−4.27	0.25
<i>L</i> = 3	14	−3.42	0.20	983	−3.50	0.20
<i>L</i> = 4	12	−2.98	0.10	323	−2.97	0.10
<i>L</i> = 5	24	−2.57	0.17	441	−2.55	0.16
<i>L</i> = 6	23	−2.15	0.07	154	−2.14	0.07
<i>L</i> = 7	47	−1.75	0.15	213	−1.77	0.15
<i>L</i> = 8	48	−1.24	0.14	143	−1.27	0.14
<i>L</i> = 9	46	−0.83	0.11	85	−0.86	0.10
<i>L</i> = 10	48	−0.46	0.12	49	−0.47	0.11
<i>L</i> = 11	47	−0.01	0.11	35	−0.04	0.15
<i>L</i> = 12	49	0.53	0.22	16	0.56	0.17
<i>L</i> = 13	45	1.16	0.20	8	1.14	0.17
<i>L</i> = 14	39	1.99	0.33	3	1.80	0.17
<i>L</i> = 15	8	3.70	1.09	0	—	—

$$Y_{ij} = \gamma_0 + \delta_z Z_{ij} + \gamma_1 (\text{Logit_}\hat{q})_{ij} + \sum_{s=2}^{15} \alpha_s L_{sij} \\ + \gamma_2 (\text{Dur_F})_{ij} + u_{0j} + u_{1j} Z_{ij} + e_{ij}, \\ \begin{pmatrix} u_{0j} \\ u_{1j} \end{pmatrix} \sim N \left[\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} \tau_{00} & \tau_{01} \\ \tau_{10} & \tau_{11} \end{pmatrix} \right]; \\ e_{ij} \sim N(0, \sigma^2). \tag{4}$$

Here L_{sij} , $s = 2, \dots, 15$, are dummy indicators for fourteen of the fifteen propensity strata that sub-classified the at-risk students. We made additional adjustment for $(\text{Logit_}\hat{q})_{ij}$, the logit of a student's estimated propensity of being retained in a retention school. In a multilevel educational system, a child's potential outcome values may depend on the treatment setting. We used u_{0j} and u_{1j} to denote the respective setting-specific

TABLE 9
Within-Stratum Mean Differences in Reading Achievement in Retention Schools

Stratum	<i>Z</i> = 1, <i>D</i> = 1			<i>Z</i> = 0, <i>D</i> = 1			Mean diff
	<i>N</i>	<i>M</i>	<i>SD</i>	<i>N</i>	<i>M</i>	<i>SD</i>	
<i>L</i> = 1	9	47.41	11.73	3,044	59.22	11.64	−11.80
<i>L</i> = 2	12	47.25	15.50	1,654	55.62	11.30	−8.37
<i>L</i> = 3	14	42.66	15.92	977	52.49	11.95	−9.83
<i>L</i> = 4	12	45.24	17.32	321	52.45	12.53	−7.21
<i>L</i> = 5	24	44.07	14.43	440	50.05	12.15	−5.97
<i>L</i> = 6	23	36.63	13.88	153	48.77	12.67	−12.14
<i>L</i> = 7	47	37.41	10.92	211	45.89	12.40	−8.48
<i>L</i> = 8	48	39.79	15.37	143	46.00	13.37	−6.21
<i>L</i> = 9	45	37.78	12.27	85	43.28	11.52	−5.50
<i>L</i> = 10	48	35.00	9.77	49	42.18	13.34	−7.18
<i>L</i> = 11	47	34.30	9.05	35	40.45	13.08	−6.16
<i>L</i> = 12	49	32.06	9.31	16	33.03	14.33	−0.96
<i>L</i> = 13	45	31.09	11.96	8	35.27	8.25	−4.18
<i>L</i> = 14	38	33.87	9.15	3	40.47	8.60	−6.60
<i>L</i> = 15	8	32.84	7.71	0	—	—	—

TABLE 10
Within-Stratum Mean Differences in Math Achievement in Retention Schools

Stratum	Z = 1, D = 1			Z = 0, D = 1			Mean diff
	N	M	SD	N	M	SD	
L = 1	9	38.88	9.22	3,044	45.56	7.58	-6.67
L = 2	12	40.87	11.14	1,654	43.48	7.70	-2.61
L = 3	14	35.12	12.65	976	41.59	8.48	-6.48
L = 4	12	35.62	12.68	321	41.77	8.50	-6.15
L = 5	24	35.64	9.20	440	39.42	8.44	-3.78
L = 6	23	31.05	9.70	153	39.02	9.41	-7.98
L = 7	47	33.37	9.44	211	37.19	9.70	-3.81
L = 8	48	34.49	10.43	143	36.97	10.09	-2.48
L = 9	45	34.98	10.95	85	34.43	9.17	0.55
L = 10	48	30.76	8.07	49	34.56	10.37	-3.80
L = 11	47	30.97	9.56	35	35.63	8.98	-4.66
L = 12	49	28.34	9.53	16	26.77	12.71	1.57
L = 13	45	27.04	10.24	8	30.96	8.22	-3.92
L = 14	38	30.10	8.58	3	37.45	5.66	-7.36
L = 15	8	27.49	11.53	0	—	—	—

increments to a child’s learning outcome and to the child-specific retention effect in each subject area. In particular, the above model enabled us to estimate the variation of the retention effect across retention schools, denoted by τ_{11} , as well as the covariance between the school-specific potential outcome of promotion and the school-specific retention effect, τ_{10} .

Results for the reading outcome are summarized in Table 11. If an at-risk child promoted in a retention school were retained instead, we estimated that his or her expected reading achievement would be 9.01 points lower at the end of the treatment year, with a standard error of 0.68 and a 95% confidence interval of (-10.42, -7.76). The magnitude of this estimated effect amounted

to about two-thirds of a standard deviation of the reading outcome. There was a statistically significant variation in the retention effect across the retention schools ($\hat{\tau}_{11} = 18.83, \chi^2 = 280.30, df = 230, p < .05$). If we assume that the retention effect is normally distributed over these schools, the effect on reading would range from -17.52 to -0.50 among 95% of the retention schools. The correlation between school-specific retention effect and school mean outcome was -0.27, indicating a more severe negative effect of kindergarten retention in retention schools in which the average reading achievement was higher.

We reported in Table 12 the estimated effect of kindergarten retention on the math learning of at-risk children attending retention schools. If an at-risk child was retained rather than being pro-

TABLE 11
Model-Based Estimation of Kindergarten Retention Effect on Reading in Retention Schools

Fixed effect		Coefficient	SE	t
Retention school promoted at-risk kid intercept, γ		53.99	0.28	192.15
Retention effect in retention schools, δ_z		-9.01	0.68	-13.27
Random effect	Variance	df	χ^2	p value
School mean intercept, u_j	55.02	230	1,863.46	.000
School retention effect, Δ_{z,u_j}	18.83	230	280.30	.013
Correlation between u_j and Δ_{z,u_j}			-0.27	
Level-1 effect, e_{ij}	88.22			

Note. For brevity, we have omitted in this table the coefficient estimates for the covariates including the propensity strata, the logit of propensity score, and the duration of treatment up to the time of assessment.

TABLE 12
Model-Based Estimation of Kindergarten Retention Effect on Mathematics in Retention Schools

Fixed effect		Coefficient	SE	t
Retention school promoted at-risk kid intercept, γ		42.32	0.19	227.95
Retention effect in retention schools, δ_z		-5.89	0.50	-11.79
Random effect	Variance	df	χ^2	p value
School mean intercept, u_j	23.32	230	1408.06	.000
School retention effect, $\Delta_{z,uj}$	12.77	230	289.14	.005
Correlation between u_j and $\Delta_{z,uj}$			0.40	
Level-1 effect, e_{ij}	44.08			

Note. For brevity, we have omitted in this table the coefficient estimates for the covariates including the propensity strata, the logit of propensity score, and the duration of treatment up to the time of assessment.

moted in a retention school, the math achievement of this child was expected to be 5.89 points lower with a standard error of 0.50. This estimated deficiency, with a 95% confidence interval of (-6.97, -5.01), was also about two-thirds of a standard deviation of the math outcome. The retention effect on the math outcome varied across the retention schools ($\hat{\tau}_{11} = 12.77$,

$\chi^2 = 289.14$, $df = 230$, $p < .01$), ranging from -12.89 to 1.11 among 95% of the schools. The correlation between school-specific retention effect and school mean outcome was 0.40, indicating that kindergarten retention had a more severe negative effect in retention schools in which the average math achievement was lower.

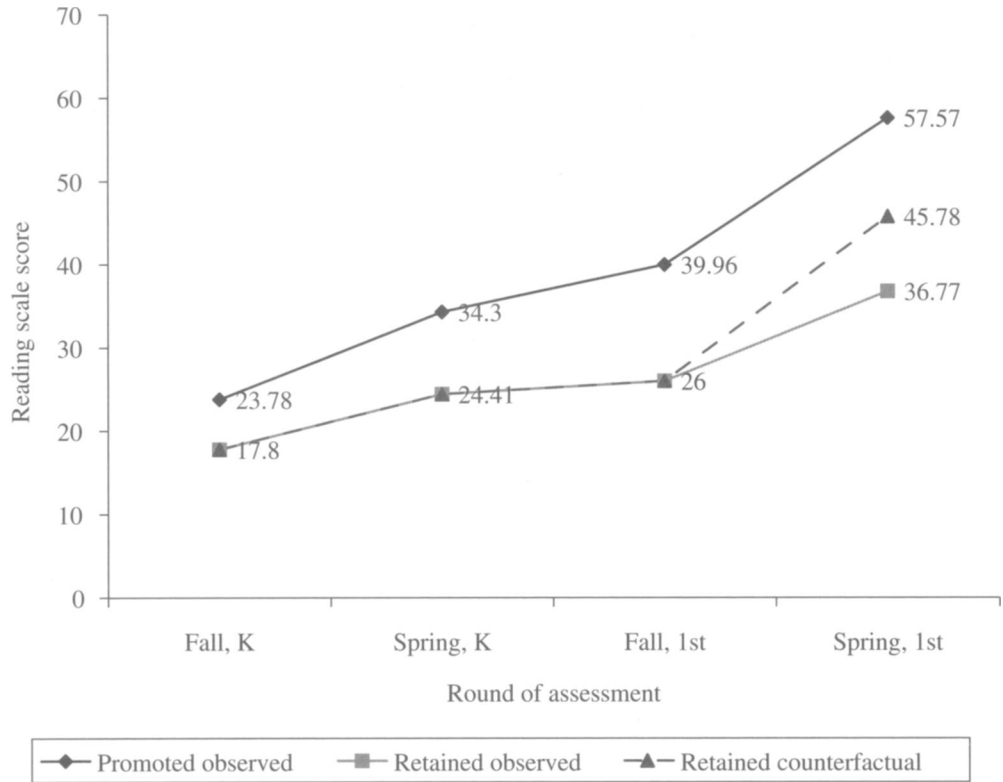


FIGURE 2. *Observed and counterfactual reading growth trajectories of retention school students.*

Figure 2 illustrates the observed reading growth trajectories of the retained group and the promoted group in retention schools. We excluded from the retained group the eight retainees who were most likely to be retained. On the basis of the above estimation results, the dashed line represents the counterfactual growth trajectory that would have been observed under our model had the kindergarten retainees been promoted in the retention schools. After a year of repetition, although children in the retained group are expected to reach the average end-of-kindergarten achievement level of their promoted peers, the gap in reading achievement between these two groups increased from about 10 points to almost 21 points over the treatment year, apparently because of the lower growth rate of the retained group. According to our analytic results, had the retainees been promoted to the first grade instead, they would have demonstrated a growth rate in reading comparable to that of their promoted peers, which would have led to a considerable reduction of the achievement gap between the two groups.

The corresponding growth trajectories in mathematics are displayed in Figure 3. Similar to the result in reading, a same-grade comparison would suggest that, after spending an additional year in kindergarten, the retainees would achieve at the same level, on average, as their promoted peers did a year before. Under our model, had the kindergarten retainees been promoted, however, the math achievement gap between the two groups by the end of the treatment year would have remained almost the same as that prior to the treatment, rather than being doubled in width.

Results from further analyses suggested that the kindergarten retention effects depended on a child's probability of being retained. This was true for both reading (coefficient of interaction = 1.73, $SE = 0.44$, $t = 3.95$) and mathematics (coefficient of interaction = 0.98, $SE = 0.36$, $t = 2.72$). The effects of kindergarten retention tended to diminish as the propensity to be retained increased. However, no inference could be made for those retainees who were most likely to be

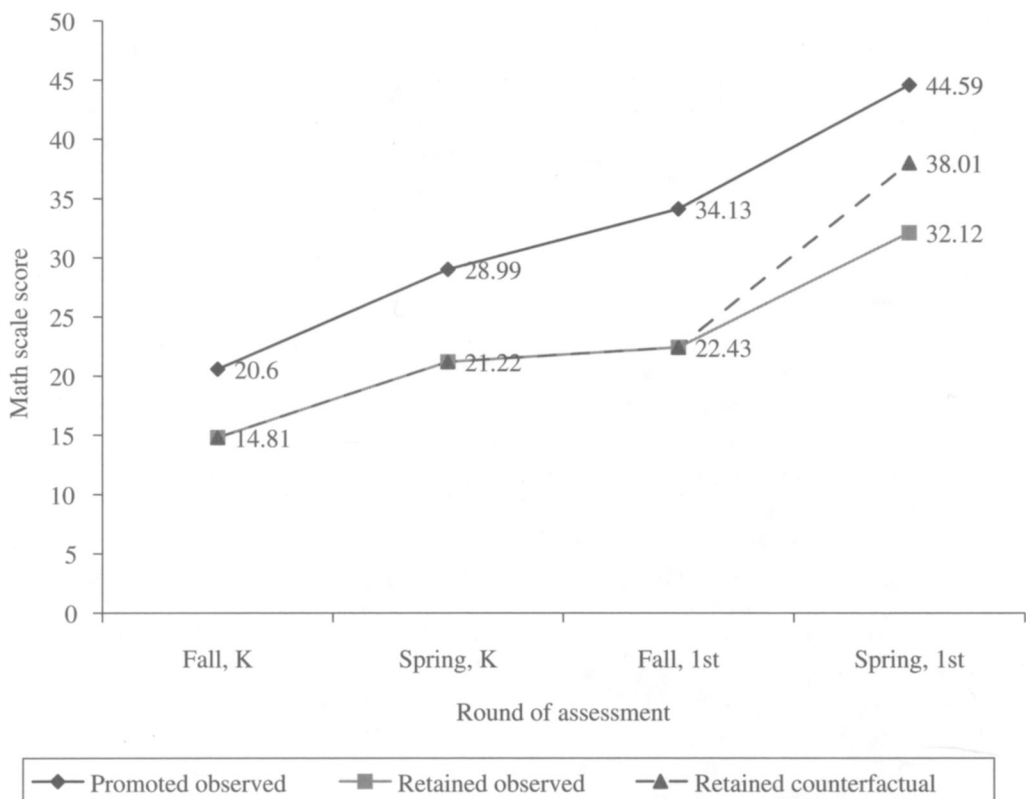


FIGURE 3. Observed and counterfactual math growth trajectories of retention school students.

retained, because very few promoted students in the retention schools could be matched to those highest-risk children, weakening statistical precision for this comparison.

Under strong ignorability, the observed average learning outcomes of the at-risk children promoted in retention schools are unbiased estimates of the expected counterfactual outcomes of the kindergarten retainees who had the same propensity score. We examined the sensitivity of the estimated retention effects to any possible departure from the strong ignorability assumption. The purpose was to see if the general conclusion would be altered by additional adjustment for some unmeasured confounders comparable to the most important covariates observed (Lin, Psaty, & Kronmal, 1998; Rosenbaum, 1986, 2002). We assumed that there might exist a student-level unmeasured composite and a school-level unmeasured composite comparable to the most important student-level and school-level observed covariates in each subject area. We then obtained bounds for the new estimates of the retention effects with adjustment for these hypothetical confounders as well as for the estimated propensity score. The 95% confidence intervals for the upper bounds of the re-adjusted retention effect estimates did not contain zero or any positive values. Therefore, our hypothesis testing results regarding the negative effects of kindergarten retention on at-risk children attending retention schools did not seem sensitive to omission of unmeasured confounders.

Conclusion and Discussion

The current study was intended to inform policy makers and educational practitioners with new evidence regarding the effects of the school-level retention policy at the kindergarten level. Although causal inference is challenging, the exceptional richness of the ECLS-K data and the propensity score-based adjustment method enabled us to achieve a relatively more comprehensive control of the pretreatment differences than previously possible.

According to our analytic results, the average effect of the kindergarten retention policy, as compared to a policy that banned retention, was null or very small. Nor did we find any evidence that the policy would benefit those children who

would be promoted if the policy were adopted. These results cast doubt on the proposition that a policy of grade retention in kindergarten would improve instruction by creating classrooms that are more homogenous in academic ability.

We did find evidence, however, that children who were retained would have learned more had they been promoted. This was true in both reading and math. After being retained for one year, the average loss in academic growth experienced by the retainees was about two-thirds of a standard deviation of the outcome in each subject area, equivalent to almost half a year's expected growth. There was evidence of a diminishing retention effect as a child's risk of retention increased; however, even for those who tended to be diagnosed as in a relatively higher need of repeating a grade, there was no evidence that they received any immediate benefit from the retention treatment. In general, kindergarten retention seemed to have constrained the learning potential of all but the highest-risk children.

The reader may wonder whether our three results are mutually consistent. If retention hurts those retained while not helping those who would have been promoted, one would suspect that, on average, the retention policy would have a negative effect. However, we found no average effect of the retention policy. The explanation can be found in that only 4.6% of the kindergartners were retained. Despite the fact that the negative impact of retention for those students was quite large, when applied to such a small fraction of students, the effect is not sufficient to produce a statistically detectable mean difference between retention schools and nonretention schools.

Summarizing the above empirical findings, we conclude that the kindergarten retention treatment leaves most retainees even further behind, and, therefore, impedes these children's cognitive development over the repetition year. In general, at-risk children promoted to the next grade level seemed to have a better chance of growth acceleration. The results from this study challenge the practice of routinely using kindergarten retention as a solution to the difficulties experienced by young children. Rather than forcing these children to restart from the very beginning, exposing them to meaningful intellectual chal-

lenges on a continual basis is perhaps developmentally more appropriate.

Quantity and quality of learning opportunities are generally viewed as major school-related causes of students' academic achievement (Carroll, 1963). Some previous research has hinted that the retention policy might show some effectiveness if schools provided the at-risk children with an increased quantity and an improved quality of learning opportunities. For example, Roderick and her colleagues (Roderick, Nagaoka, Bacon, & Easton, 2000) detected some positive effects of Chicago's Summer Bridge program on the academic growth of children who were at the verge of being retained. However, the retention policy may not be a necessary condition for children to benefit from better learning opportunities. According to Smith and Shepard's (1988) report, some schools were able to adapt instructional arrangements to at-risk children's needs without enforcing the retention policy; in contrast, instructional adaptation across grade levels was rare in schools with a relatively large proportion of retainees.

Some results from this study will direct us to further investigation that may inform the above debate. We found a significant variation of the retention effect in each subject area across the retention schools. This may be attributed to the variation in schools' and teachers' enactment of the retention and the promotion treatments, the variation in student groups' responses to the treatments, and the interaction between these factors. In particular, a more severe effect of kindergarten retention in reading was found in schools with a higher average reading achievement; whereas, a more severe retention effect in mathematics was displayed in schools with a lower average math achievement. We suspect that curricular and instructional differences between kindergarten and first grade were more drastic in these two types of schools. Reading instruction typically progresses with a faster pace in high-achieving schools, perhaps at the expense of the retained children. Meanwhile, schools with a lower average math achievement usually introduce very little math content in kindergarten. In both cases, repeating the kindergarten curriculum seemed to have placed the retainees at further disadvantage. More evidence is in need regarding whether and

how the retention effects depend on individualized adaptation and instructional coordination across grade levels. The ECLS-K data set contains information about curricular content and instructional activities delivered at the class level and about special services provided to individual students during the treatment year. Such information can be exploited in future research for disclosing the mechanisms through which the learning opportunities moderate or mediate the retention effects.

A widely accepted argument for kindergarten retention is that, after learning the same material for the second time, the retained children may gain edge and bloom in the later years. The same-grade comparisons, as shown in Figures 2 and 3, suggest that the relative standing of the kindergarten retainees was generally favorable when compared with children from a younger cohort. However, previous research has indicated that such academic benefits were temporary (Jimerson et al., 1997), and that retained children were more likely to drop out from secondary schools (Grissom & Shepard, 1989; Jimerson, 1999; Rumberger & Larson, 1998). Future studies need to examine whether the at-risk children would be equally likely to drop out if they had been promoted instead. The current data set does not enable us to estimate the long-term effects of the kindergarten retention treatment on children's school success and career accomplishment. However, future waves of data collection in ECLS-K will allow empirical study of this question.

The current study is focused on the effects of kindergarten retention on children's cognitive growth in reading and mathematics. Our results say nothing about whether the kindergarten retainees developed a higher self-esteem or were stigmatized among their younger peers. The retention effects on children's emotional outcomes are beyond the scope of this study. Nonetheless, the causal inference theories and methods can be applied to address questions about long-term cognitive and emotional effects of grade retention.

In conclusion, the empirical evidence from this study refutes the arguments that adopting a kindergarten retention policy boosts achievement on average, that such a policy improves the learning of children who would in any case be promoted, or that grade retention helps children experiencing difficulty in kindergarten. School-

to-school variation in the magnitude of the impact of retention suggests that how schools treat the retained and promoted children will moderate the negative effect of retention. However, the range of plausible effects across schools does not include any important positive effects of retention. Meanwhile, the negative effects in many schools are quite large. A principle of “doing no harm” applied to these results undermines support for kindergarten retention.

Notes

¹In principle, we could attempt to answer this question for all children at risk of repetition, including those who attend schools that ban retention. But this requires a good estimate of the probability that each student who attends a school that bans retention would be retained if that school allowed retention. Such an estimate can be justified only under very strong assumptions about the similarity in practices between schools that do and do not ban retention. We, therefore, focus only on those children who attend schools that allow retention when we study the average causal effect of being retained.

²The purpose of the propensity model is not to allow a parsimonious prediction model for the retention policy but rather to create strata that are balanced on pretreatment covariates. In large samples, if two treatment groups within a stratum are balanced on the propensity score, they will be balanced on all the covariates used to predict propensity (Rosenbaum & Rubin, 1983). In practice, with finite samples, balance must be checked empirically (Rosenbaum & Rubin, 1984; Rubin, 1997). It is in the interest of the investigator to use any and all pretreatment covariates, including nonlinear terms, to ensure balance.

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